

Partnership

When two or more persons join hands for a business to attain profits is called **Partnership** and the persons are called **partners**.

Types of Partners :-

1. Active Partners: The partner who manages the business is known as Active Partner or working partner.

2. Sleeping Partner: A partner, who simply invests his money in the business but does not manage it voluntarily is referred as Sleeping partner.

Every partner invests some amount of money for a certain time to help the partnership firm to get profits.

Types of partnership :-

1. Simple partnership – If all partners invest their different amounts of capital (money) for the same period or the same capital for a different period, then such a partnership is called a simple partnership.

2. Compound partnership – If all the partners invest their different capitals (money) for different periods, then such a partnership is called a compound partnership.

In Compound partnership partners' profit not only depends on their investments but also on the time period.

Case 1. When capital investment is the same for the different time periods :-

$$P_1 : P_2 : P_3 = T_1 : T_2 : T_3$$

Ex.1 - Aftab, Tarun and Aditya started a business by investing Rs. 8000 each. But after six months Aftab left and after 8 months Tarun left. If at the end of the year, there was a gain of Rs. 7800, then what will be the share of Tarun?

Sol. Ratio of Profit

$$\rightarrow \text{Aftab} : \text{Tarun} : \text{Aditya} = 6 : 8 : 12$$

$$\text{Share of Tarun} \rightarrow \frac{8}{26} \times 7800 = 2400 \text{ Rs.}$$

Case 2. When capital investment X_1, X_2 and X_3 is different for the different time period T_1, T_2 and T_3 .

$$P_1 : P_2 : P_3 = X_1 T_1 : X_2 T_2 : X_3 T_3$$

Ex.2 - Aman and Bhanu invest some amount in the ratio 4 : 5 in a business. After 8 months, Chandan joins them with an investment equal to thrice that of Bhanu. In what ratio should the yearly profit be shared among Aman, Bhanu

and Chandan?

Sol.

Ratio of profit is equal to the product of investment and time of investment.

Ratio $\rightarrow$ Aman : Bhanu : Chandan

Investment $\rightarrow$ 4 : 5 : 15

Time $\rightarrow$ 12 : 12 : 4

Profit $\rightarrow$ 48 : 60 : 60 = 4 : 5 : 5

Ex.3 - Rajesh and Sanjay are partners in a business firm. Rajesh puts in Rs. 5000 and Sanjay puts in Rs. 4000. Rajesh receives 10 % of the profit for managing the business as he is the active partner and the rest of the profit is divided in the ratio of their capitals. If the total profit is Rs. 1000, find Rajesh's total share.

Sol.

Here the investment time is the same, then the profit will be shared in the ratio of their investment.

Rajesh gets 10% of total profit as he is

$$\text{active partner} \rightarrow 1000 \times \frac{10}{100} = 100 \text{ Rs.}$$

Remaining Profit = 900 Rs.

Ratio of Profit $\rightarrow$ 5000 : 4000 = 5 : 4

Rajesh total share $\rightarrow$ 100 + 500 = 600 Rs.

Variety Questions

Q.1. Mayur and Aham enter into a partnership by making investments in the ratio 2 : 3. 10% of the total profit goes to charity. If Aham's share is ₹4,860, then what is the total profit (in ₹) earned?

SSC MTS 11/09/2023 (3rd Shift)

(a) 7,200 (b) 9,000 (c) 8,500 (d) 11,000

Q.2. Nikita made two investments; she earns 10% on one investment and lost 20% on the second investment. If the investments were in the ratio 2 : 3, then she got a:

SSC MTS 05/09/2023 (2nd Shift)

(a) loss of 8% (b) profit of 15%
(c) profit of 8% (d) loss of 15%

Q.3. A lady invests ₹27,600 in three shares in the ratio 3 : 4 : 5 which pay dividends of 20%, 15% and 5% on her investment for that year, respectively. What is the total dividend?

SSC MTS 04/09/2023 (1st Shift)

(a) ₹3,852 (b) ₹3,335
(c) ₹3,125 (d) ₹2,535

Q.4. Swati starts a business with ₹4,200 and after 5 months, Bani joins Swati as her partner. After a year, the profit is divided among Swati and Bani in the ratio 2 : 5. What is Bani's contribution?

SSC MTS 01/09/2023 (3rd Shift)

(a) ₹18,000 (b) ₹17,050
(c) ₹19,000 (d) ₹16,500

Q.5. Three persons A, B and C invested some money in a business. The share of A is twice the share of B, and the share of B is thrice the share of C. After 1 year, they earned ₹6,000. Find the money A received.

SSC CHSL 11/08/2023 (4th Shift)

(a) ₹3,200 (b) ₹3,000
(c) ₹3,600 (d) ₹3,500

Q.6. Three partners P, Q and R invest ₹1,000, ₹1,200 and ₹1,500, respectively, in a business. What should be the share of Q if the total profit is ₹666?

SSC CHSL 10/08/2023 (2nd Shift)

(a) ₹216 (b) ₹275 (c) ₹252 (d) ₹284

Q.7. Kapil initiated a business with an investment of ₹30,00,000 in April 2022. After six months, his friend Kunal joined him with an investment of ₹25,00,000. In February 2023, Kapil's Ex-Class fellow Saloni also invested an amount equal to the total investment made by the earlier two partners. In what ratio the profit earned after 3 years from the date on which Kapil initiated the business, will be distributed?

SSC CHSL 02/08/2023 (4th Shift)

(a) 332 : 200 : 50 (b) 108 : 75 : 143
(c) 432 : 300 : 55 (d) 432 : 400 : 55

Q.8. A is a working partner and B is a sleeping partner in a business. A invested ₹30,000 and B invested ₹50,000. A receives 20% of the profit for managing the business and the rest is divided in proportion to their capital. What does B get out of the profit of ₹10,000?

SSC CHSL 02/08/2023 (2nd Shift)

(a) ₹5,500 (b) ₹5,000
(c) ₹4,500 (d) ₹4,000

Q.9. X, Y and Z enter into a partnership. X invests some money at the beginning. Y invests thrice the amount of X after 4 months and Z invest double the amount of Y after 9 months from the beginning. If the annual gain be Rs. 450000, then what is the share of Y?

SSC MTS 08/05/2023 (Morning)

(a) Rs. 200000 (b) Rs. 150000
(c) Rs. 100000 (d) Rs. 300000

Q.10. P invested Rs. 4000 for one year. Q joined him with an investment of Rs. 6000 after a few months. If at the end of one year the ratio of profits of P and Q is 4 : 3, then Q joined after how many months?

SSC CHSL 14/03/2023 (1st Shift)

(a) 4 months (b) 3 months
(c) 8 months (d) 6 months

Q.11. A, B and C started a business with initial investments of ₹ 20000, ₹25000 and ₹10000, respectively. After 5 months from start, A invested ₹ 4000 more. After 6 months from start, C invested ₹ 8000 more. After 4 months from start, B withdrew ₹ 8000. At the end of the year, they will receive a profit of ₹ 'x'. In what ratio will they share the profits?

SSC CGL Tier II (08/08/2022)

- (a) 71 : 57 : 42 (b) 71 : 59 : 42
(c) 59 : 68 : 42 (d) 67 : 59 : 42

Q.12. A, B and C started a business with their capitals in the ratio 2 : 3 : 5. A increased his capital by 50% after 4 months, B increased his capital by $33\frac{1}{3}$

% after 6 months and C withdrew 50% of his capital after 8 months, from the start of the business. If the total profit at the end of a year was Rs. 86,800, then the difference between the shares of A and C in the profit was :

SSC CGL Tier II (13/09/2019)

- (a) Rs.12,600 (b) Rs.7,000
(c) Rs.9,800 (d) Rs.8,400

Q.13. A, B and C started a business by investing Rs 1,37,500 and Rs 1,62,500 and Rs 1,87,500 respectively. A is a working partner and gets 20% of the profit as working allowance and remaining is distributed in the proportion of their investment. If the total profit is Rs 2,19,375. What is the share of C?

SSC CPO 13/03/2019 (Evening)

- (a) Rs 88,600 (b) Rs 64,500
(c) Rs 67,500 (d) Rs 62,700

Practice Questions

SSC CHSL 2024 Tier - 2

Q.14. A, B and C jointly started a business in which A invested ₹7,000 for 12 months, B invested ₹8,000 for 10 months and C invested ₹12,000 for 6 months. A was the working member for which he received 20% of the profits as his remuneration. The total profit earned at the end of one year was ₹14,750. What is C's share (in ₹) in the profit?

SSC CHSL Tier II (18/11/2024)

- (a) 4,000 (b) 2,950
(c) 3,600 (d) 4,200

SSC CGL 2024 Tier - 1

Q.15. A, B and C invested money in a business such that 2 times A's share, 4 times B's share and 6 times C's share are all equal. If they earned ₹44,000, find the share of A.

SSC CGL 24/09/2024 (1st Shift)

- (a) ₹22,000 (b) ₹12,000
(c) ₹24,000 (d) ₹20,000

SSC Selection Post (Phase - XII)

Q.16. Three partners decided to start a business. They decided to donate 10% of the profit to a charity organisation and then the remaining profit would be shared among first, second and third partner in the ratio 1 : 2 : 2. The total profit received was ₹1,80,000. Find the share of the first partner.

Matriculation Level 26/06/2024 (Shift - 1)

- (a) ₹32,400 (b) ₹31,800
(c) ₹33,200 (d) ₹29,600

SSC CHSL 2023 Tier - 2

Q.17. A and B enter into a partnership with ₹25,000 and ₹50,000, respectively, in a company for a year. After 7 months, C gets into a partnership with them with ₹30,000 and A withdraws his contribution after 9 months. What would be C's share in the profit, assuming a total profit of ₹3,900 at the end of the year?

SSC CHSL Tier II (02/11/2023)

- (a) ₹600 (b) ₹800 (c) ₹1,200 (d) ₹1,000

SSC MTS 2023 Tier - 1

Q.18. Arvind, Bilal, Carmen invest ₹49,000 towards a business. Arvind invests ₹5,000 more than Bilal and Bilal invests ₹4,000 more than Carmen. Out of a total profit of ₹98,000, how much does Bilal receive?

SSC MTS 11/09/2023 (1st Shift)

- (a) ₹34,400 (b) ₹33,200
(c) ₹30,500 (d) ₹32,000

Q.19. A, B and C started a business. A's capital is 7 times the capital of C and B's capital is 2 times the capital of A. If the profit is ₹7,920, then find the share of C (in ₹).

SSC MTS 13/09/2023 (3rd Shift)

- (a) 350 (b) 360 (c) 365 (d) 355

SSC CHSL 2023 Tier - 1

Q.20. Anam and Bikash invest in a business in the ratio 3 : 2. If in a particular month, Anam gets ₹3,840 as profit, the total profit is:

SSC CHSL 04/08/2023 (2nd Shift)

- (a) ₹5,800 (b) ₹6,300
(c) ₹6,550 (d) ₹6,400

Q.21. A, B and C started a business by investing ₹9,000, ₹18,000 and ₹ 21,000, respectively. If B's share in the profit

earned by them is ₹759, then the total profit earned by them together is:

SSC CHSL 10/08/2023 (3rd Shift)

- (a) ₹2,808 (b) ₹2,024
(c) ₹2,504 (d) ₹2,208

SSC CGL 2023 Tier - 1

Q.22. A and B invested money in a business in the ratio of 7 : 5. If 15% of the total profit goes for charity, and A's share in the profit is Rs.5,950, then what is the total profit?

SSC CGL 18/07/2023 (4th shift)

- (a) Rs.12,500 (b) Rs.12,000
(c) Rs.10,500 (d) Rs.11,750

SSC Selection Post (Phase - XI)

Q.23. A starts a business with ₹45,000 and was joined afterwards by B with ₹30,000. When did B join if the profit at the end of the year was divided in the ratio 2 : 1 ?

Matriculation Level 30/06/2023 (Shift - 4)

- (a) After 6 months (b) After 12 months
(c) After 3 months (d) After 9 months

SSC CHSL 2022 Tier - 2

Q.24. A and B jointly start a business. The investment of A is equal to seven times the investment of B. Find the share of A (in ₹) in the annual profit of ₹2,75,000.

SSC CHSL Tier II (26/06/2023)

- (a) 2,40,625 (b) 2,45,625
(c) 2,42,365 (d) 2,41,225

SSC MTS 2022 Tier - 1

Q.25. Rs.1200 are divided among X, Y and Z such that Y's share is Rs.200 more than X's share. Y's share is also Rs.200 less than Z's share. The share of Y is:

SSC MTS 12/05/2023 (Evening)

- (a) Rs.200 (b) Rs.600
(c) Rs.800 (d) Rs.400

Q.26. A and B started a business with Rs.15000 and Rs.30000 respectively. If A and B share profit in the ratio of 1: 5 respectively, then what is the ratio of the time period of investment for A and B respectively ?

SSC MTS 11/05/2023 (Afternoon)

- (a) 1 : 2 (b) 2 : 5 (c) 2 : 3 (d) 2 : 7

Q.27. Three partners P, Q and R invested a total of Rs. 52,000 in a business. At the end of the year, P got Rs. 1430, Q got Rs.1870 and R got Rs. 2420 as the share of profit. How much amount did Q invest

in the business ?

SSC MTS 13/06/2023 (Evening)

- (a) Rs. 17000 (b) Rs. 20400
(c) Rs. 15300 (d) Rs. 18700

SSC CHSL 2022 Tier - 1

Q.28. P, Q and R are partners and invest in a business. P invests $\frac{1}{5}$ th of the total investments and Q invests $\frac{1}{4}$ th of the total investments. What is the ratio of investments of P, Q and R respectively?

SSC CHSL 13/03/2023 (1st Shift)

- (a) 4 : 8 : 11 (b) 4 : 5 : 11
(c) 4 : 7 : 11 (d) 4 : 5 : 9

Q.29. X, Y and Z invested Rs.4,000, Rs.6,000 and Rs.8,000 respectively in a business. They invested the money for 3, 2 and 4 years respectively. If the total profit is Rs.56,000, then what is the share of Y in the profit?

SSC CHSL 15/03/2023 (2nd Shift)

- (a) Rs.18,000 (b) Rs.12,000
(c) Rs.15,000 (d) Rs.9,000

Q.30. Nikhil and Neha invested capital in the ratio 5 : 4. The time Neha invested her money for was one year more than that invested by Nikhil. The ratio of their profits is in the ratio 6 : 5. Find the time Neha invested the money for.

SSC CHSL 21/03/2023 (1st Shift)

- (a) 26 years (b) 23 years
(c) 24 years (d) 25 years

SSC CGL 2022 Tier - 2

Q.31. A, B and C did certain investments and the ratio of their time periods is 3 : 2 : 7 respectively. Ratio of the profits of A, B and C is 4 : 3 : 14 respectively. What is the ratio of the investments of A, B and C ?

SSC CGL Tier II (02/03/2023)

- (a) 1 : 3 : 4 (b) 7 : 9 : 11
(c) 8 : 9 : 12 (d) 2 : 3 : 11

Q.32. Suresh, Dinesh and Ramesh became partners in a business by investing money in the ratio of 3 : 6 : 8. If their investments are increased by 5%, 15% and 20%, respectively, then what will be the ratio of their profits for one year ?

SSC CGL Tier II (03/03/2023)

- (a) 7 : 46 : 64 (b) 19 : 46 : 64
(c) 21 : 46 : 64 (d) 35 : 46 : 64

Q.33. A starts a business with ₹75,000 and B joins the business 5 months later with an investment of ₹80,000. After 1 year, they earn a profit of ₹4,08,800. Find the share of A and B (in ₹).

SSC CGL Tier II (07/03/2023)

- (a) 2,52,000 and 1,56,800
(b) 2,50,000 and 1,58,800
(c) 2,48,000 and 1,60,800
(d) 2,49,500 and 1,59,300

SSC CGL 2021 Tier - 2

Q.34. A, B and C started a business with the investment of ₹ 100000, ₹ 140000, ₹ 200000 respectively. After 3 months, C left the business, 7 months after C left the business, B also left the business. B and C took their investment with them. At the end of the year, C received his share of profit as ₹ 1155. What is the total share of profits of A and B ?

SSC CGL Tier II (08/08/2022)

- (a) ₹ 6150 (b) ₹ 5005
(c) ₹ 4995 (d) ₹ 5555

SSC MTS 2020 Tier - 1

Q.35. The profit earned by a company is to be divided among three friends who invested their money in the company, in the proportion of 6 : 9 : 11. If the share of the person who invested the most is ₹2,310, then what is the sum of the money (in ₹) received by the other two friends?

SSC MTS 22/10/2021 (Morning)

- (a) 1,890 (b) 3,150 (c) 5,460 (d) 1,260

SSC CGL 2020 Tier - 1

Q.36. Keshav, Surjeet and Thomas started a business with investments in the ratio 2 : 3 : 4. The ratio of their period of investments is 5 : 6 : 9. Twenty percent of the profit was spent on rent and maintenance of the office. Remaining profit was distributed among themselves. If the difference in the share of profit of Keshav and Surjeet is Rs.7264 then how much is the total profit (in Rs.)?

SSC CGL 18/08/2021 (Evening)

- (a) 51060 (b) 58112 (c) 46490 (d) 72640

SSC CPO 2020 Tier - 1

Q.37. A, B, C subscribe a sum of Rs.75,500 for a business. A subscribes Rs.3,500 more than B, and B subscribes Rs. 4,500 more than C. Out of a total profit of Rs.45,300 how much (in Rs.) does A receive ?

SSC CPO 25/11/2020 (Evening shift)

- (a) 12,600 (b) 15,000
(c) 17,400 (d) 14,700

Answer Key :-

1.(b)	2.(a)	3.(b)	4.(a)
5.(c)	6.(a)	7.(b)	8.(b)
9.(a)	10.(d)	11.(d)	12.(a)
13.(c)	14.(c)	15.(c)	16.(a)
17.(a)	18.(d)	19.(b)	20.(d)
21.(b)	22.(b)	23.(c)	24.(a)
25.(d)	26.(b)	27.(a)	28.(b)
29.(b)	30.(d)	31.(c)	32.(c)
33.(a)	34.(b)	35.(b)	36.(d)
37.(c)			

Solutions :-

Sol.1.(b) Let the total profit is ₹x

According to the question,

$$x \times \frac{9}{10} \times \frac{3}{5} = ₹4860 \Rightarrow x = ₹9,000$$

Sol.2.(a)

Let the investment 200 and 300 units

According to the question,

$$200 \times \frac{11}{10} = 220 \text{ units}$$

$$300 \times \frac{4}{5} = 240 \text{ units}$$

Total investment = 200 + 300 = 500 units

Total earning = 220 + 240 = 460 units

$$\text{Loss \%} = \frac{500 - 460}{500} \times 100 = 8 \%$$

Sol.3.(b) Amount invested in first share

$$= 27,600 \times \frac{3}{12} = 6900$$

Amount invested in second share

$$= 27,600 \times \frac{4}{12} = 9200$$

Amount invested in third share

$$= 27,600 \times \frac{5}{12} = 11500$$

$$\text{Total dividend} = 6900 \times \frac{1}{5} + 9200 \times \frac{3}{20}$$

$$+ 11500 \times \frac{1}{20} = ₹3,335$$

Sol.4.(a) Let Bani's contribution is x

$$4200 \times 12 : x \times 7 = 2 : 5$$

$$x = \frac{4200 \times 12 \times 5}{7 \times 2} = ₹18000$$

Sol.5.(c) Let the investment of C = Rs. x
B's investment = Rs. 3x then A's investment = Rs. 6x

Since, they invest their money for 1 year,

Their profit ratio = 6 : 3 : 1

Hence, money received by A

$$= \frac{6}{10} \times 6000 = \text{Rs. } 3,600$$

Sol.6.(a) Investment time is the same ,

So ratio of investment is equal to the ratio of profit .

Their investment ratio :-

$$P : Q : R = 10 : 12 : 15$$

$$\text{Share of Q} = 666 \times \frac{12}{37} = 18 \times 12 = ₹216$$

Sol.7.(b) Ratio = kapil : kunal : saloni

$$\text{Amount} = 30 : 25 : 55$$

$$\text{Time} = 36 : 30 : 26$$

$$\text{Profit} = 108 : 75 : 143$$

Sol.8.(b) A receive 20% of the profit for managing the business.

Remaining profit excluding A's managing profit = $10,000 \times 80\% = 8,000$

$$\text{B's profit} = 8,000 \times \frac{5}{8} = 5,000 \text{ Rs.}$$

Sol.9.(a)

$$\text{Ratio} \rightarrow X : Y : Z$$

$$\text{profit} \rightarrow x \times 12 : 3x \times 8 : 6x \times 3$$

$$\text{Total profit} = 12x + 24x + 18x = 54x$$

$$54x = ₹450,000$$

$$\text{Share of Y (24x)} = \frac{450,000}{54x} \times 24x = ₹200,000$$

Sol.10.(d)

$$\text{Ratio} \rightarrow P : Q$$

$$\text{Investment} \rightarrow 4 : 6$$

$$\text{Time} \rightarrow 1 : x$$

$$\text{Profit} \rightarrow 4 : 3$$

Let Q joined for x year

As we know profit = investment $\times$ time

$$\text{Then, } 4 \times 1 : 6 \times x = 4 : 3 \Rightarrow \frac{4}{6x} = \frac{4}{3}$$

$$\Rightarrow x = \frac{3}{6} = \frac{1}{2}$$

Here, Q joined for 6 months then, Q joined after $(12 - 6) = 6$ month

Sol.11.(d) Total amount invested by A

$$= 20,000 \times 5 + (20,000 + 4,000) \times 7$$

$$= 1,00,000 + 1,68,000 = 2,68,000$$

Total amount invested by B

$$= 25,000 \times 4 + (25,000 - 8,000) \times 8$$

$$= 1,00,000 + 1,36,000 = 2,36,000$$

Total amount invested by C

$$= 10,000 \times 6 + (10,000 + 8,000) \times 6$$

$$= 60,000 + 1,08,000 = 1,68,000$$

So, the ratio of their profit

$$= 268 : 236 : 168 = 67 : 59 : 42$$

$$\text{Sol.12.(a)} \quad 50\% = \frac{1}{2}, \quad 33\frac{1}{3}\% = \frac{1}{3}$$

$$\text{Ratio} \rightarrow A : B : C$$

$$\text{Initial} \rightarrow 2 \times 4 : 3 \times 6 : 5 \times 8$$

$$\text{Final} \rightarrow 3 \times 8 : 4 \times 6 : 2.5 \times 4$$

$$\text{Profit} \rightarrow 32 : 42 : 50$$

$$\text{Total Profit } (32 + 42 + 50) = 124 \text{ units}$$

$$124 \text{ units} = 86,800 \text{ Rs.}$$

$$1 \text{ unit} = 700 \text{ Rs.}$$

Difference b/w share of A and C $\rightarrow$

$$(50 - 32) \text{ units} = 18 \text{ units}$$

$$18 \text{ units} = 18 \times 700 = 12,600 \text{ Rs.}$$

Sol.13.(c)

$$\text{Ratio} \rightarrow A : B : C$$

$$\text{Investment} \rightarrow 137,500 : 1,62,500 : 1,87,500$$

$$11 : 13 : 15$$

Therefore, C's share :-

$$2,19,375 \times \frac{4}{5} \times \frac{15}{39} = \text{Rs. } 67,500$$

Sol.14.(c)

$$\text{Share} \rightarrow A : B : C$$

$$\text{Investment} \rightarrow 7,000 : 8,000 : 12,000$$

$$\text{Time} \rightarrow 12 : 10 : 6$$

$$\text{Ratio} \rightarrow 84 : 80 : 72$$

$$21 : 20 : 18$$

A received 20% of the profits as his remuneration

$$\text{Remaining profit} = 14,750 \times \frac{4}{5} = 11,800$$

$$59 \text{ unit} = ₹11,800$$

$$\text{C's share (18 unit)} = \frac{11,800}{59} \times 18 = ₹3,600$$

Sol.15.(c) If 2 times A's share, 4 times B's share and 6 times C's share are all equal

$$\therefore 2A = 4B = 6C$$

$$\text{Hence, } A : B : C = 6 : 3 : 2$$

$$\text{Now, A's share} = \frac{6}{11} \times 44,000 = ₹24,000$$

Sol.16.(a)

Total profit distributed among partners in business = $1,80,000 \times 90\% = ₹1,62,000$

Share of the 1st partner

$$= \left(\frac{1}{1+2+2} \right) \times 1,62,000$$

$$= \frac{1,62,000}{5} = ₹32,400$$

Sol.17.(a)

Ratio of their money contribution

$$A : B : C = 5 : 10 : 6$$

Ratio of their time

$$A : B : C = 9 : 12 : 5$$

$$\therefore \text{Ratio of their profit} = 45 : 120 : 30$$

$$= 3 : 8 : 2$$

Hence,

$$\text{C's share in profit} = \frac{3,900}{13} \times 2 = 600$$

Sol.18.(d) Let investment of Bilal be x

According to the question,

$$x + (x + 5,000) + (x - 4,000) = 49,000$$

$$3x = 48,000, x = 16,000$$

Then, investment of Bilal : Arvind : Carmen

$$= 16,000 : 21,000 : 12,000$$

$$= 16 : 21 : 12$$

Share of profit of Bilal

$$= \frac{98,000}{16 + 21 + 12} \times 16 = ₹32,000$$

Sol.19.(b) Concept :- profit is divided in the ratio of capital

$$\text{Ratio of profit of A, B, and C} = 7 : 14 : 1$$

$$\text{Share of C} = 7,920 \times \frac{1}{(7 + 14 + 1)} = 360$$

Sol.20.(d) 3 units = 3,840 ₹

$$\text{Total profit (5 units)} = \frac{3,840}{3} \times 5 = 6,400 ₹$$

Sol.21.(b) Their investment time is same, so the ratio of profit is equal to the ratio of investment.

Ratio of investment :-

$$A : B : C = 3 : 6 : 7$$

$$\text{B's share of profit (6 units)} = ₹759$$

$$\text{Total profit (16 units)} = \frac{759}{6} \times 16$$

$$= 253 \times 8 = ₹2,024$$

Sol.22.(b) Let the investment of A and B be 7x and 5x.

If investment time is equal, then the ratio of profit is equal to the ratio of investment.

According to the question,

$$7x = 5,950 \text{ Rs.}$$

$$12x = 5,950 \times \frac{12}{7} = 10,200 \text{ Rs.}$$

$$\text{Total profit} = 10,200 \times \frac{100}{(100 - 15)} = 12,000 \text{ Rs.}$$

Sol.23.(c)

$$\text{Ratio} \rightarrow A : B$$

$$\text{Investment} \rightarrow 3 : 2$$

$$\text{Time} \rightarrow 12 : x$$

$$\text{Profit} \rightarrow 2 : 1$$

x (time for which B invested)

$$= \frac{12 \times 3}{2 \times 2} = 9 \text{ month}$$

Required time after which B invested

$$= 12 - 9 = 3 \text{ months}$$

Sol.24.(a) If investment time is equal, then the ratio of profit is equal to the ratio of investment.

$$\text{Ratio} \rightarrow A : B$$

$$\text{Invest} \rightarrow 7 : 1$$

$$\text{Profit} \rightarrow 7 : 1$$

$$\text{Share of A} = 2,75,000 \times \frac{7}{8} = 2,40,625 ₹$$

Sol.25.(d)

$$\text{Ratio} \rightarrow X : Y : Z$$

$$\text{Amount} \rightarrow a : a + 200 : a + 400$$

$$\text{Total amount} = 3a + 600$$

$$3a + 600 = ₹1,200$$

$$a = \frac{1,200 - 600}{3} = ₹200$$

$$\text{Share of Y} = a + 200 = 200 + 200 = ₹400$$

Sol.26.(b)

$$\text{Ratio} \rightarrow A : B$$

$$\text{Investment} \rightarrow 1 : 2$$

$$\text{Time} \rightarrow x : y$$

$$\text{Profit} \rightarrow 1 : 5 \Rightarrow \frac{1x}{2y} = \frac{1}{5}$$

Required Ratio of time period = $x : y = 2 : 5$

Sol.27.(a) As we know :-

profit = time $\times$ investment

If investment time is equal, then the ratio of profit is equal to the ratio of investment.

According to the question,

Total profit of P, Q and R = Rs. 5720

$$\text{Investment of Q} = \frac{1870}{5720} \times 52000 \\ = \text{Rs. } 17000$$

Sol.28.(b) Let the total investment = 100

Then, Investment of P = 20 and investment of Q = 25

So, investment of R = $(100 - 45) = 55$

Therefore, ratio of investment of (P : Q : R) = 20 : 25 : 55 $\Rightarrow 4 : 5 : 11$

Sol.29.(b)

Ratio $\rightarrow X : Y : Z$

Amount $\rightarrow 2 : 3 : 4$

Time $\rightarrow 3 : 2 : 4$

profit $\rightarrow 6 : 6 : 16$

$$\text{the share of Y} = 56,000 \times \frac{6}{28} \\ = \text{Rs } 12000$$

Sol.30.(d)

Ratio $\rightarrow$ Nikhil : Neha

Investment $\rightarrow 5 : 4$

Time $\rightarrow x : x + 1$

Profit $\rightarrow 6 : 5$

Now,

$$\frac{5x}{4(x+1)} = \frac{6}{5} \Rightarrow 25x = 24x + 24 \\ \Rightarrow x = 24 \text{ years}$$

So Neha invested for $(x + 1)$ years = 25 years

Sol.31.(c) Let, the investments of A, B and C be x, y, z respectively.

Total Investment $\times$ Total time = Profit

Ratio $\rightarrow A : B : C$

Profit $\rightarrow 3x : 2y : 7z = 4 : 3 : 14$

$$\Rightarrow x : y : z = \frac{4}{3} : \frac{3}{2} : \frac{14}{7}$$

= 8 : 9 : 12

Sol.32.(c)

Ratio $\rightarrow$ Suresh : Dinesh : Ramesh

Investment $\rightarrow 300 : 600 : 800$

Increased investment $\rightarrow$

315 : 690 : 960 or 21 : 46 : 64

Sol.33.(a) Profit = Investment $\times$ Time

$$75000 \times 12 : 80000 \times 7 \Rightarrow 45 : 28$$

73 units = ₹ 4,08,800

45 units = ₹ 2,52,000

28 units = ₹ 1,56,800

Therefore Profit of A and B will be ₹ 2,52,000 and ₹ 1,56,800 respectively.

Sol.34.(b) Ratio of investment of A : B : C = 100000 : 140000 : 200000 = 5 : 7 : 10

According to the question,

A works for 12 months, B works for 3 + 7

= 10 months, C works for 3 months.

Profit sharing ratio among A : B : C =

$$(5 \times 12) : (7 \times 10) : (10 \times 3) = 30 : 35 : 15$$

Let total profit be y .

$$\text{Now, } \frac{15}{80} \times y = 1155 \Rightarrow y = ₹ 6,160$$

So, total profit of A and B

$$= \frac{65}{80} \times 6160 = ₹ 5,005$$

Sol.35.(b) Ratio of investment = 6 : 9 : 11.

Let the shares of profit be

6x, 9x and 11x.

According to the question,

$$11x = 2310 \Rightarrow x = 210.$$

The shares of the other two

$$= 6x + 9x = 15x = 15 \times 210 = 3,150.$$

Sol.36.(d)

Investments $\rightarrow 2 : 3 : 4$

Time period $\rightarrow 5 : 6 : 9$

Profit $\rightarrow 10 : 18 : 36$

According to question,

Profit difference of Keshav and Surjeet $\rightarrow$

$$(18 - 10) \text{ units} = 8 \text{ units} = 7264 \text{ Rs.}$$

1 unit = 908 Rs.

Total Profit $(10 + 18 + 36) = 64$ units

64 units = 58112 Rs.

Let, total profit = 100%

Profit shared = 80%

$$80\% \rightarrow 58112 \text{ Rs.} \Rightarrow 100\% \rightarrow 72640 \text{ Rs.}$$

Sol.37.(c)

Total sum of A, B and C = 75,500 Rs.

$$B = C + 4500$$

$$A = B + 3500 \text{ or } C + 8000$$

According to the question

$$(C + 8000) + (C + 4500) + C = 75,500$$

$$3C + 12,500 = 75,500$$

$$C = 21,000 \text{ Rs.}, A = 29,000 \text{ Rs.}$$

$$\text{Share of A} = \frac{29,000}{75,500} \times 45,300$$

$$= 2900 \times 6 = 17,400 \text{ Rs.}$$